

Geometry Quest

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

What an Angle Is

Every answer is a number of degrees, or a count. Type digits only.

Warm-Up 8 ITEMS • RECALL

1 Right angles in a full turn

2 Right angles in a straight line

3 Perimeter of a 5 by 2 rectangle

4 Perimeter of a 4 by 4 square

5 Perimeter of a 8 by 3 rectangle

6 Perimeter of a 7 by 7 square

7 Perimeter of a 9 by 1 rectangle

8 Perimeter of a 2 by 2 square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. Perimeter of a 14 by 6 rectangle

.....

10. A square of perimeter 24 — one side

.....

11. A rectangle of perimeter 26, one side 8 — the other

.....

12. Perimeter of a 20 by 11 rectangle

.....

13. A square of perimeter 64 — one side

.....

★

Challenge

NOT GRADED · REASON IT OUT

14. Degrees in a right angle
-
15. Degrees in a straight line
-
16. Degrees all the way round
-
17. Degrees in half a right angle
-
18. On a straight line: 130° and $?^\circ$ (180 - 130)
-
19. Round a point: 90° , 120° and $?^\circ$ (360 - 210)
-
20. Three angles of a triangle: 60° , 70° and $?^\circ$ (They add to 180)
-
21. An angle of 200° — how far past a straight line? (200 - 180)
-
22. Two equal angles on a straight line — each is $?^\circ$ (180 shared evenly)
-
23. Angles in a square, added up (Four right angles)
-
24. Turn 90° four times — total degrees (All the way round)
-
25. A rectangle of perimeter 24 with whole sides — how many different ones
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Perimeter & Area

Perimeter is the fence. Area is the grass. Read which one is being asked for.

Warm-Up 12 ITEMS • RECALL

1 Perimeter of a 5 by 3 rectangle

2 Area of a 5 by 3 rectangle

3 Perimeter of a square with side 6

4 Area of a square with side 6

5 Perimeter of a 10 by 2 rectangle

6 Area of a 10 by 2 rectangle

7 Area of a 5 by 2 rectangle

8 Area of a 4 by 4 square

9 Area of a 8 by 3 rectangle

10 Area of a 7 by 7 square

11 Area of a 9 by 1 rectangle

12 Area of a 6 by 5 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area 48, one side 6 — the other side ($48 \div 6$)

.....

14. Perimeter 30, one side 9 — the other side ($9 + 9$ is 18, and 12 is left for two sides)

.....

15. Area of a 12 by 7 rectangle (84 square units)

.....

16. Perimeter of a 12 by 7 rectangle ($12 + 12 + 7 + 7$)

.....

17. L-shape: a 6 by 4 block plus a 3 by 2 block — total area ($24 + 6$)

.....

18. Area of a 14 by 6 rectangle

.....

19. Area 60, one side 5 — the other

.....

20. Area of a 20 by 11 rectangle

.....

21. A square of area 81 — one side

.....

22. Area 169 with equal sides — one side

.....

★

Challenge

NOT GRADED · REASON IT OUT

23. Biggest area with perimeter 24 (whole sides) (The 6 by 6 square)
-
24. Smallest area with perimeter 24 (whole sides) (1 by 11)
-
25. Area of an L: a 7 by 7 square minus a 3 by 2 corner
-
26. Double both sides of a 3 by 6 rectangle — the new area
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

The Coordinate Plane

Across first, then up. (3, 5) is a different place from (5, 3).

Warm-Up 12 ITEMS • RECALL

1 In (3, 5), the across number

2 In (3, 5), the up number

3 In (7, 2), the across number

4 In (0, 4), the across number

5 In (6, 6), the up number

6 In (9, 1), the up number

7 A 1 by 9 rectangle — perimeter

8 That rectangle — area

9 A 5 by 5 square — perimeter

10 That square — area

11 A 2 by 8 rectangle — area

12 A 3 by 7 rectangle — area

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Distance from (2, 3) to (7, 3) (Only the across number changed)

14. Distance from (4, 1) to (4, 8) ($8 - 1$)

15. Rectangle at (1,1), (6,1), (6,4), (1,4) — its perimeter (5 wide and 3 tall)

16. That same rectangle — its area (5×3)

17. Start at (2,2), go 4 across and 3 up — the up number now ($2 + 3$)

18. With 20 m of fence, the biggest whole-number area

19. With 28 m of fence, the biggest area

20. With 12 m of fence, the biggest area

21. With 20 m of fence, the smallest area using whole sides

22. With 36 m of fence, the biggest whole-number area

★ Challenge NOT GRADED • REASON IT OUT

23. Square with corners (0,0), (5,0), (5,5) — the fourth corner's up number (It sits at (0, 5))

24. Distance from (1,2) to (1,9) plus (3,4) to (8,4) (7 + 5)

25. Among rectangles with perimeter 32, the biggest area

26. With 100 m of fence, the biggest whole-number area

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Sorting Shapes & Symmetry

Sort by what is true about a shape, not by what it looks like.

Warm-Up 12 ITEMS • RECALL

1 Sides on a hexagon

2 Sides on a pentagon

3 Right angles in a rectangle

4 Equal sides on an equilateral triangle

5 Sides on an octagon

6 Pairs of parallel sides in a rectangle

7 A 4 by 2 plus a 4 by 2 — total area

8 A 5 by 5 plus a 3 by 3

9 A 6 by 3 plus a 6 by 1

10 A 7 by 2 plus a 4 by 2

11 A 12 by 1 plus a 6 by 2

12 A 8 by 3 plus a 2 by 3

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Lines of symmetry in a square (Two through the middle, two through the corners)

.....

14. Lines of symmetry in a rectangle (not a square) (The corner folds don't match)

.....

15. Lines of symmetry in an equilateral triangle (One from each corner)

.....

16. Lines of symmetry in a circle — type 0 if too many to count (Infinitely many, so type 0)

.....

17. A 9 by 7 with a 3 by 2 removed — area

.....

18. A 12 by 12 with a 5 by 5 removed

.....

19. An L of a 10 by 5 and a 6 by 4

.....

20. A 14 by 6 with a 4 by 6 removed

.....

21. A T of a 10 by 2 and a 2 by 7

.....

★ Challenge NOT GRADED • REASON IT OUT

22. Degrees in a quarter turn (A right angle)

23. Lines of symmetry in a regular hexagon (One per pair of opposite corners and sides)

24. Sides of a shape whose angles add to 360 (Any quadrilateral)

25. A 25 by 18 room with a 5 by 6 alcove added — total area

26. A 15 by 15 with a 15 by 4 strip removed

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Angle Guess & Map a Planet

Enrichment day. Estimate before you measure, every single time.

✦ Warm-Up 2 ITEMS • RECALL

1 With 20 m and no wall,
the biggest area

2 Sides you must fence
with a wall behind you

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 20 m against a wall, 5 m deep each side — the width

.....

4. That pen's area

.....

Challenge NOT GRADED • REASON IT OUT

- Angle on a straight line next to 47° (180 - 47)
- Angles round a point: 100° , 130° and $?^\circ$ (360 - 230)
- Rectangle perimeter 20 with the largest area — that area (The 5 by 5 square)
- Rectangle area 36 with the smallest perimeter — that perimeter (6 by 6)
- Distance from (3,7) to (11,7) (11 - 3)
- A shape with 4 lines of symmetry and 4 equal sides — its angle count (It's a square)
- Turn 45° eight times — total degrees (45×8)
- Interior angles of a triangle, added (Always)
- 20 m against a wall, the biggest area
- 32 m against a wall, the biggest area
- How much bigger is that than the no-wall best for 32 m

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • What an Angle Is

OUT OF 13

Warm-Up • 1-8 • 4 | 2 | 14 | 16 | 22 | 28 | 20 | 8

Core • 9-13 • 40 | 6 | 5 | 62 | 16

Challenge • 14-25 • 90 | 180 | 360 | 45 | 50 | 150 | 50 | 20 | 90 | 360 | 360 | 6

1.2 • Perimeter & Area

OUT OF 22

Warm-Up • 1-12 • 16 | 15 | 24 | 36 | 24 | 20 | 10 | 16 | 24 | 49 | 9 | 30

Core • 13-22 • 8 | 6 | 84 | 38 | 30 | 84 | 12 | 220 | 9 | 13

Challenge • 23-26 • 36 | 11 | 43 | 72

1.3 • The Coordinate Plane

OUT OF 22

Warm-Up • 1-12 • 3 | 5 | 7 | 0 | 6 | 1 | 20 | 9 | 20 | 25 | 16 | 21

Core • 13-22 • 5 | 7 | 16 | 15 | 5 | 25 | 49 | 9 | 9 | 81

Challenge • 23-26 • 5 | 12 | 64 | 625

1.4 • Sorting Shapes & Symmetry

OUT OF 21

Warm-Up • 1-12 • 6 | 5 | 4 | 3 | 8 | 2 | 16 | 34 | 24 | 22 | 24 | 30

Core • 13-21 • 4 | 2 | 3 | 0 | 57 | 119 | 74 | 60 | 34

Challenge • 22-26 • 90 | 6 | 4 | 480 | 165

Fri • Angle Guess & Map a Planet

OUT OF 4

Warm-Up • 1-2 • 25 | 3

Core • 3-4 • 10 | 50

Challenge • 5-15 • 133 | 130 | 25 | 24 | 8 | 4 | 360 | 180 | 50 | 128 | 64

Perimeter

The distance all the way round. Add every side, or be clever about it.

Warm-Up 12 ITEMS • RECALL

1 Perimeter of a 4 by 3 rectangle

2 Perimeter of a 5 by 5 square

3 Perimeter of a 10 by 2 rectangle

4 Perimeter of a 6 by 8 square

5 Perimeter of a 7 by 1 rectangle

6 Perimeter of a 3 by 3 square

7 Perimeter of a 6 by 2 rectangle

8 Perimeter of a 8 by 8 square

9 Perimeter of a 12 by 1 rectangle

10 Perimeter of a 9 by 9 square

11 Perimeter of a 5 by 4 rectangle

12 Perimeter of a 10 by 10 square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Perimeter of a 12 by 8 rectangle

.....

14. A square of perimeter 36 — one side

.....

15. A rectangle of perimeter 20, one side 6 — the other

.....

16. Perimeter of a 15 by 9 rectangle

.....

17. A square of perimeter 100 — one side

.....

18. Perimeter of a 16 by 9 rectangle

.....

19. A square of perimeter 48 — one side

.....

20. A rectangle of perimeter 34, one side 10 — the other

.....

21. Perimeter of a 25 by 15 rectangle

.....

22. A square of perimeter 200 — one side

.....

★ Challenge NOT GRADED • REASON IT OUT

23. A rectangle of perimeter 30 with whole sides — how many different ones

.....

24. Perimeter of an L made from a 5 by 5 square with a 2 by 2 corner removed

.....

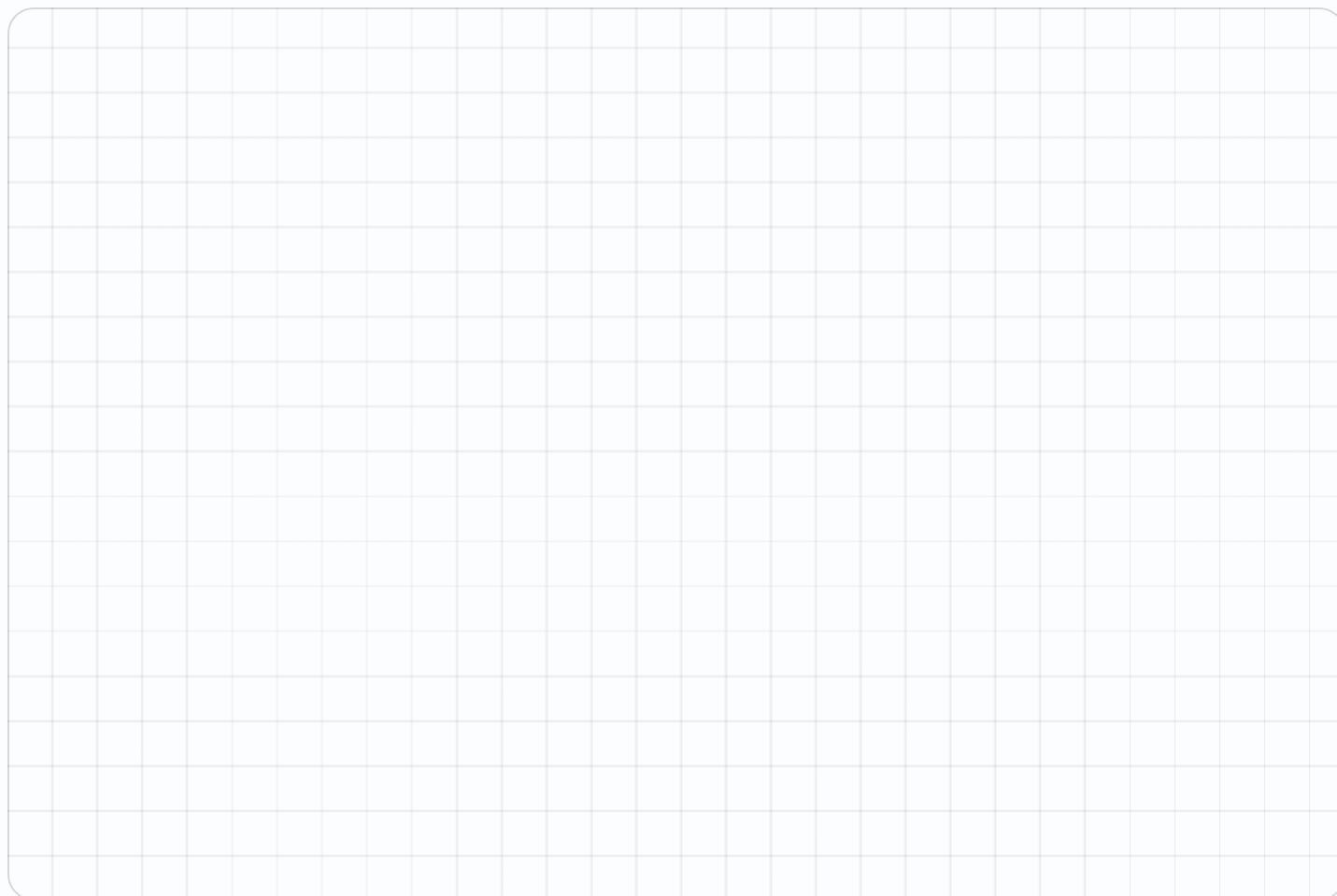
25. A rectangle of perimeter 40 with whole sides — how many different ones

.....

26. Perimeter of an L from a 8 by 8 square with a 3 by 3 corner removed

.....

Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal ONE SENTENCE PER MISTAKE

Area

Length times width — and count the squares if you doubt it.

Warm-Up 12 ITEMS • RECALL

1 Area of a 4 by 3 rectangle

2 Area of a 5 by 5 square

3 Area of a 10 by 2 rectangle

4 Area of a 6 by 6 square

5 Area of a 7 by 1 rectangle

6 Area of a 8 by 3 rectangle

7 Area of a 6 by 2 rectangle

8 Area of a 8 by 8 square

9 Area of a 12 by 1 rectangle

10 Area of a 9 by 9 square

11 Area of a 5 by 4 rectangle

12 Area of a 10 by 7 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area of a 12 by 8 rectangle

.....

14. Area 48, one side 6 — the other

.....

15. Area of a 15 by 9 rectangle

.....

16. A square of area 49 — one side

.....

17. Area 144 with equal sides — one side

.....

18. Area of a 16 by 9 rectangle

.....

19. Area 72, one side 8 — the other

.....

20. Area of a 25 by 15 rectangle

.....

21. A square of area 100 — one side

.....

22. Area 225 with equal sides — one side

.....

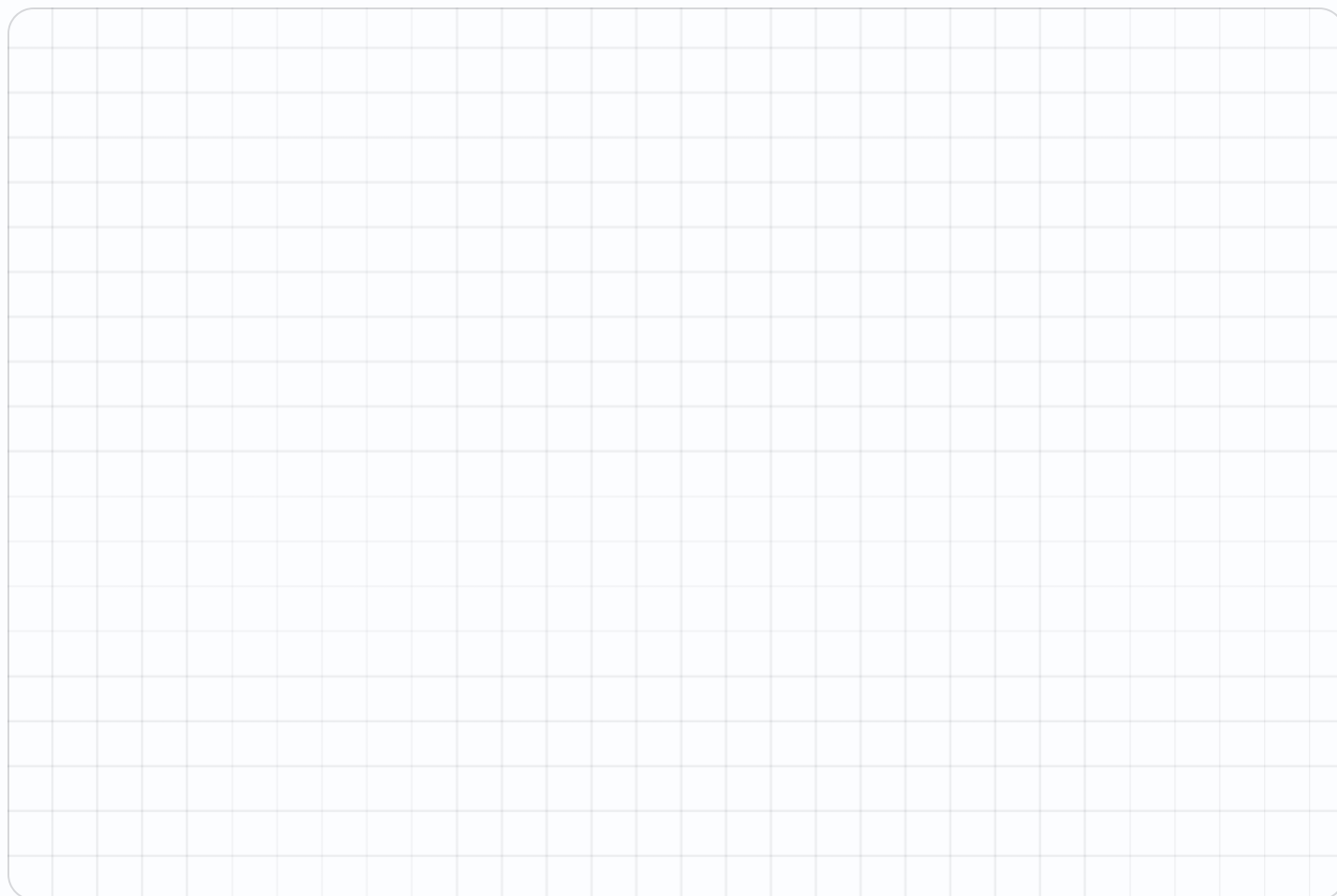
★ Challenge NOT GRADED • REASON IT OUT

23. Area of an L: a 6 by 6 square minus a 2 by 3 corner

24. Double both sides of a 4 by 5 rectangle — the new area

25. Area of an L: a 8 by 8 square minus a 3 by 4 corner

26. Double both sides of a 5 by 7 rectangle — the new area

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Same Fence, Different Field

Perimeter fixed, area wildly different. That is the Big Question.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

 Warm-Up 12 ITEMS • RECALL

1 A 1 by 11 rectangle — perimeter _____	2 That rectangle — area _____	3 A 6 by 6 square — perimeter _____	4 That square — area _____
5 A 2 by 10 rectangle — area _____	6 A 4 by 8 rectangle — area _____	7 A rectangle cut into 2 equal parts — one part is 1 over what _____	8 A rectangle cut into 3 equal parts — one part is 1 over what _____
9 A rectangle cut into 4 equal parts — one part is 1 over what _____	10 A rectangle cut into 6 equal parts — one part is 1 over what _____	11 A rectangle cut into 8 equal parts — one part is 1 over what _____	12 A 1 by 15 rectangle — perimeter _____

 Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

16. With 24 m of fence, the biggest whole-number area	_____
17. With 20 m of fence, the biggest area	_____
18. With 16 m of fence, the biggest area	_____
19. With 24 m of fence, the smallest area using whole sides	_____
20. With 30 m of fence, the biggest whole-number area	_____
21. A shape of 8 squares split into 2 equal parts — squares in each part	_____
22. A shape of 12 squares split into 4 equal parts — squares in each part	_____
23. A shape of 9 squares split into 3 equal parts — squares in each part	_____
24. A shape of 18 squares split into 6 equal parts — squares in each part	_____
25. A square folded into 4 equal parts — each part is what fraction, give the denominator	_____
26. Two of four equal parts — the same as one out of how many	_____

★ Challenge NOT GRADED • REASON IT OUT

32. The shape that always gives the biggest area for a fixed perimeter — type square or circle

33. Among rectangles with perimeter 40, the biggest area

34. A 12-square shape split into 3 equal parts, then one part split in 2 — squares in the smallest piece

35. The shape giving the biggest area for a fixed perimeter — type square or circle

36. Among rectangles with perimeter 60, the biggest area

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Area of Compound Shapes

Split it into rectangles, do each, add them up.

Warm-Up 12 ITEMS • RECALL

1 A 3 by 2 plus a 3 by 2 — total area

2 A 4 by 4 plus a 2 by 2

3 A 5 by 3 plus a 5 by 1

4 A 6 by 2 plus a 3 by 2

5 A 10 by 1 plus a 5 by 2

6 A 7 by 3 plus a 1 by 3

7 A 4 by 3 plus a 4 by 3 — total area

8 A 6 by 6 plus a 3 by 3

9 A 7 by 4 plus a 7 by 1

10 A 9 by 2 plus a 4 by 2

11 A 15 by 1 plus a 5 by 3

12 A 6 by 4 plus a 2 by 4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. An 8 by 6 with a 2 by 3 removed — area

.....

14. A 10 by 10 with a 4 by 4 removed

.....

15. An L of a 9 by 4 and a 5 by 3

.....

16. A 12 by 5 with a 3 by 5 removed

.....

17. A T of a 8 by 2 and a 2 by 6

.....

18. A 10 by 8 with a 3 by 4 removed — area

.....

19. A 14 by 14 with a 6 by 6 removed

.....

20. An L of a 12 by 5 and a 7 by 4

.....

21. A 16 by 7 with a 5 by 7 removed

.....

22. A T of a 12 by 3 and a 3 by 8

.....

★

Challenge

NOT GRADED · REASON IT OUT

23. A 20 by 15 room with a 4 by 5 alcove added — total area
-
24. A 12 by 12 with a 12 by 3 strip removed
-
25. A 30 by 20 room with a 6 by 5 alcove added — total area
-
26. A 20 by 20 with a 20 by 6 strip removed
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

The Biggest Pen

24 metres of fence, and a wall to build against. Now what?

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

- 1 With 24 m and no wall, the biggest area

- 2 Sides you must fence with a wall behind you

- 3 With 36 m and no wall, the biggest area

- 4 Fenced sides with a wall behind

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 24 m against a wall, 8 m deep each side — the width

.....
6. That pen's area

.....
7. 36 m against a wall, 9 m deep each side — the width

.....

13. How much bigger than the no-wall best for 40 m

Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • Perimeter

OUT OF 22

Warm-Up • 1-12 • 14 | 20 | 24 | 24 | 16 | 12 | 16 | 32 | 26 | 36 | 18 | 40

Core • 13-22 • 40 | 9 | 4 | 48 | 25 | 50 | 12 | 7 | 80 | 50

Challenge • 23-26 • 7 | 20 | 10 | 32

2.2 • Area

OUT OF 22

Warm-Up • 1-12 • 12 | 25 | 20 | 36 | 7 | 24 | 12 | 64 | 12 | 81 | 20 | 70

Core • 13-22 • 96 | 8 | 135 | 7 | 12 | 144 | 9 | 375 | 10 | 15

Challenge • 23-26 • 30 | 80 | 52 | 140

2.3 • Same Fence, Different Field

OUT OF 23

Warm-Up • 1-12 • 24 | 11 | 24 | 36 | 20 | 32 | 2 | 3 | 4 | 6 | 8 | 32

Core • 16-26 • 36 | 25 | 16 | 11 | 56 | 4 | 3 | 3 | 3 | 4 | 2

Challenge • 32-36 • circle | 100 | 2 | circle | 225

2.4 • Area of Compound Shapes

OUT OF 22

Warm-Up • 1-12 • 12 | 20 | 20 | 18 | 20 | 24 | 24 | 45 | 35 | 26 | 30 | 32

Core • 13-22 • 42 | 84 | 51 | 45 | 28 | 68 | 160 | 88 | 77 | 60

Challenge • 23-26 • 320 | 108 | 630 | 280

Fri • The Biggest Pen

OUT OF 7

Warm-Up • 1-4 • 36 | 3 | 81 | 3

Core • 5-7 • 8 | 64 | 18

Challenge • 8-13 • 72 | 72 | 36 | 162 | 200 | 100

Plotting Points

Across the hall, then up the stairs. Type coordinates like 3,5.

Warm-Up 12 ITEMS • RECALL

1 In (3, 5), the number across

2 In (3, 5), the number up

3 The origin — type as a,b

4 4 across, 2 up — type as a,b

5 0 across, 6 up — type as a,b

6 In (7, 1), the first number

7 In (5, 2), the number across

8 In (5, 2), the number up

9 6 across, 0 up — type as a,b

10 2 across, 7 up — type as a,b

11 0 across, 9 up — type as a,b

12 In (4, 8), the first number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 3 right of (2,5) — type as a,b

14. 2 up from (4,1) — type as a,b

15. Distance from (2,3) to (7,3)

16. Distance from (4,1) to (4,8)

17. Halfway between (0,0) and (8,0) — type as a,b

18. 5 right of (3,4) — type as a,b

19. 3 up from (6,2) — type as a,b

20. Distance from (1,5) to (9,5)

21. Distance from (7,2) to (7,10)

22. Halfway between (0,0) and (12,0) — type as a,b

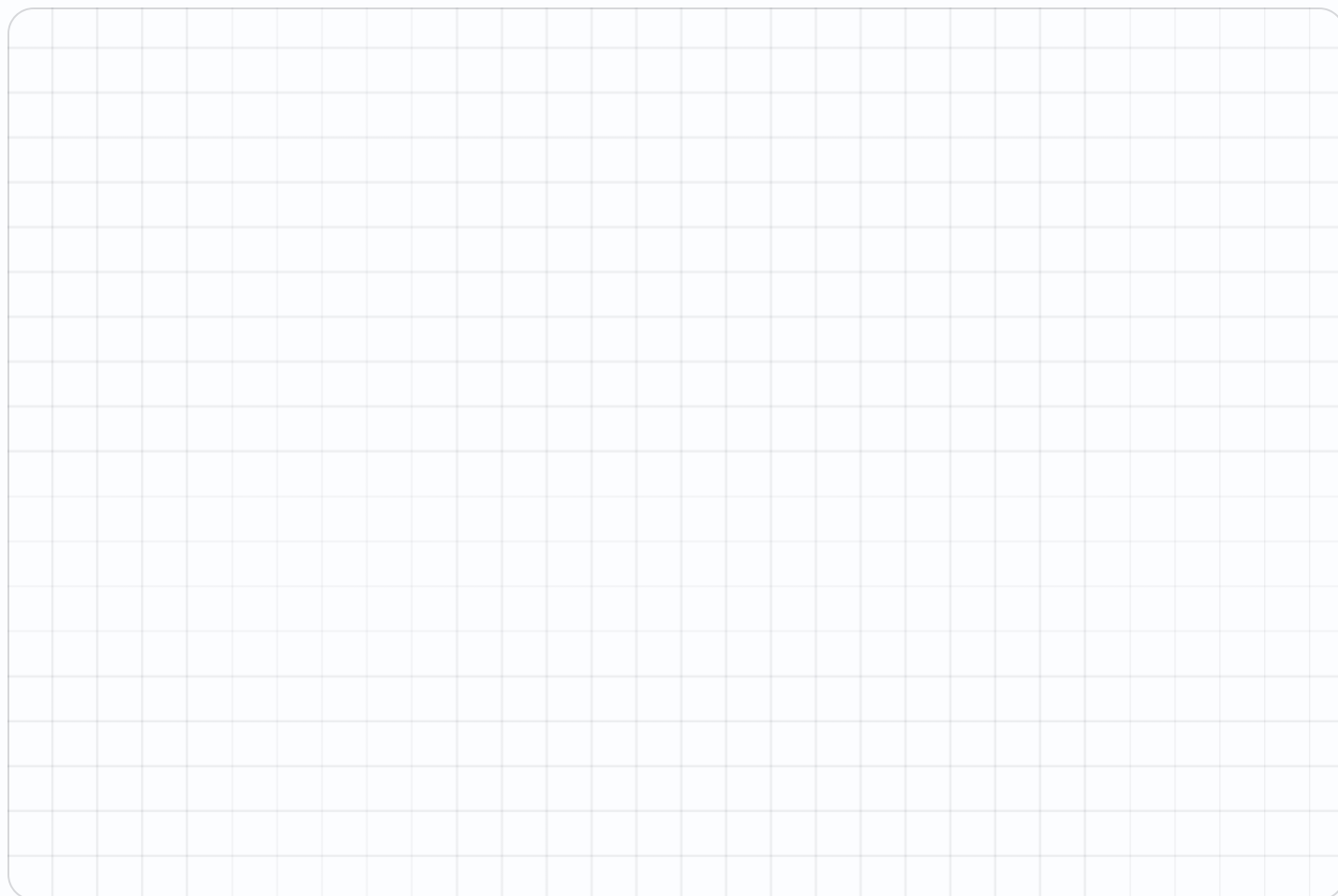
★ Challenge NOT GRADED • REASON IT OUT

23. Halfway between (2,2) and (8,6) — type as a,b

24. From (0,0) to (6,8) going only across and up — total steps

25. Halfway between (1,3) to (9,7) — type as a,b

26. From (0,0) to (7,9) going only across and up — total steps

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Shapes on the Grid

Plot the corners, then read off what you built.

Warm-Up 12 ITEMS • RECALL

1 Corners on a rectangle

2 Corners on a triangle

3 (1,1) to (5,1) — the length

4 (1,1) to (1,4) — the length

5 Sides on a square

6 (0,0) to (3,0) — the length

7 Corners on a square

8 Corners on a pentagon

9 (2,1) to (7,1) — the length

10 (2,1) to (2,6) — the length

11 Sides on a rectangle

12 (0,0) to (5,0) — the length

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (1,1), (1,5), (6,5), (6,1) — the area

.....

14. That shape's perimeter

.....

15. (2,2), (2,7), (5,7), (5,2) — the area

.....

16. (0,0), (0,4), (4,4), (4,0) — the shape's name

.....

17. The missing corner of a rectangle at (1,1),(1,4),(6,4) — type as a,b

.....

18. (1,1), (1,6), (5,6), (5,1) — the area

.....

19. (3,2), (3,8), (7,8), (7,2) — the area

.....

20. (0,0), (0,6), (6,6), (6,0) — the shape's name

.....

21. The missing corner of a rectangle at (2,2),(2,7),(8,7) — type as a,b

.....

★ Challenge NOT GRADED • REASON IT OUT

22. $(0,0)$, $(8,0)$, $(8,5)$, $(0,5)$ — the area

.....

23. A square with corners $(2,2)$ and $(7,7)$ opposite — its area

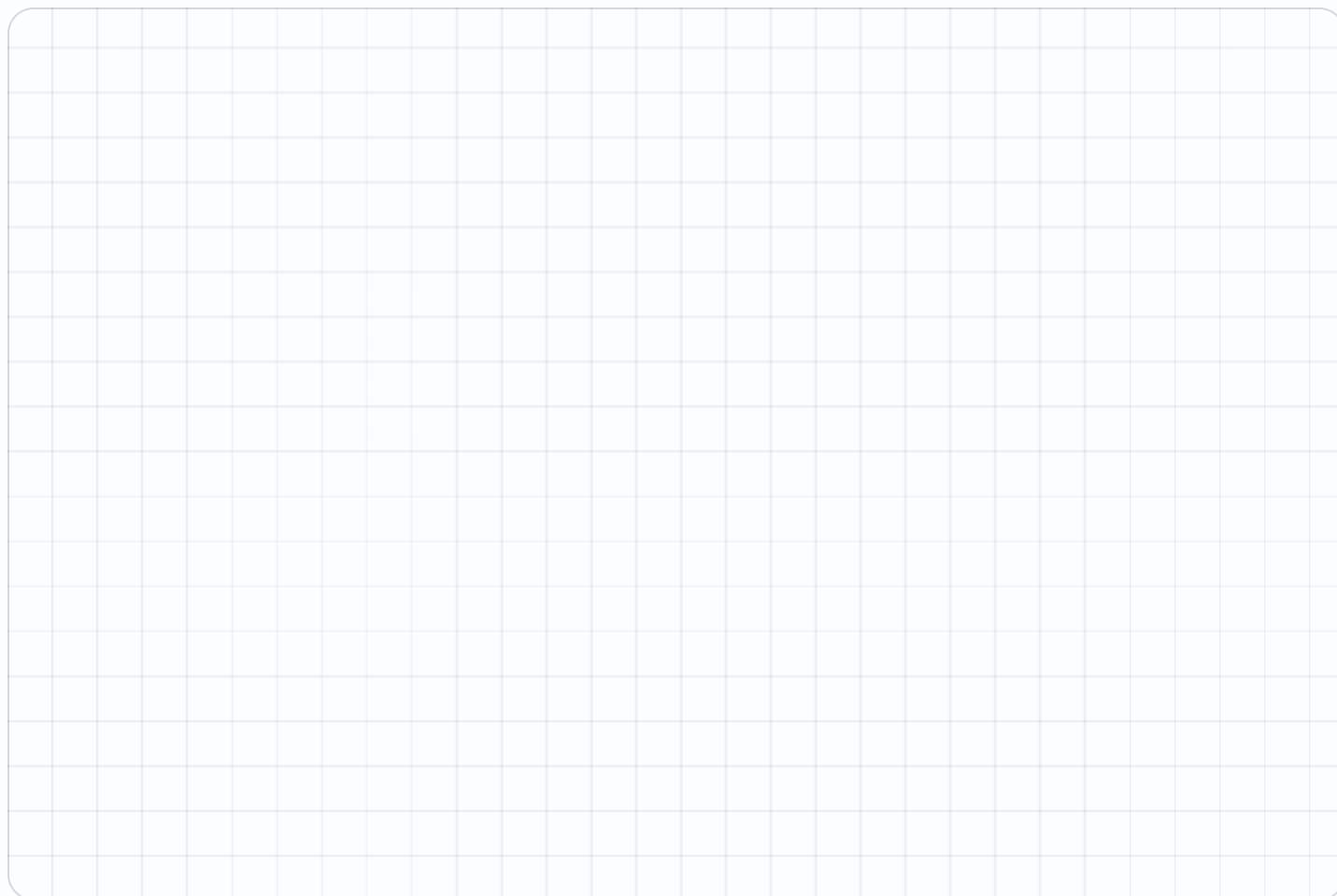
.....

24. $(0,0)$, $(10,0)$, $(10,6)$, $(0,6)$ — the area

.....

25. A square with corners $(1,1)$ and $(8,8)$ opposite — its area

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Reading a Map

Coordinates are directions somebody else has to follow.

Warm-Up 12 ITEMS • RECALL

1 From (0,0), move 3 across — type as a,b

2 From (3,0), move 2 up — type as a,b

3 From (5,5), move 1 across — type as a,b

4 From (2,2), move 2 up — type as a,b

5 Blocks from (0,0) to (4,0)

6 Blocks from (0,0) to (0,7)

7 From (0,0), move 4 across — type as a,b

8 From (4,0), move 3 up — type as a,b

9 From (6,6), move 2 across — type as a,b

10 From (3,3), move 4 up — type as a,b

11 Blocks from (0,0) to (6,0)

12 Blocks from (0,0) to (0,9)

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. From (1,2) to (6,2) then up 3 — type the end as a,b

14. Total blocks travelled on that route

15. A park at (3,4), a school at (9,4) — blocks apart

16. From (2,1) to (2,9) — blocks

17. From (0,0) to (5,5) across-and-up — blocks

18. From (2,3) to (8,3) then up 4 — type the end as a,b

19. A park at (2,5), a school at (11,5) — blocks apart

20. From (3,2) to (3,11) — blocks

21. From (0,0) to (6,6) across-and-up — blocks

★ Challenge NOT GRADED • REASON IT OUT

22. A route (1,1) to (7,1) to (7,6) — total blocks

.....

23. The shortest across-and-up route from (2,3) to (9,8) — blocks

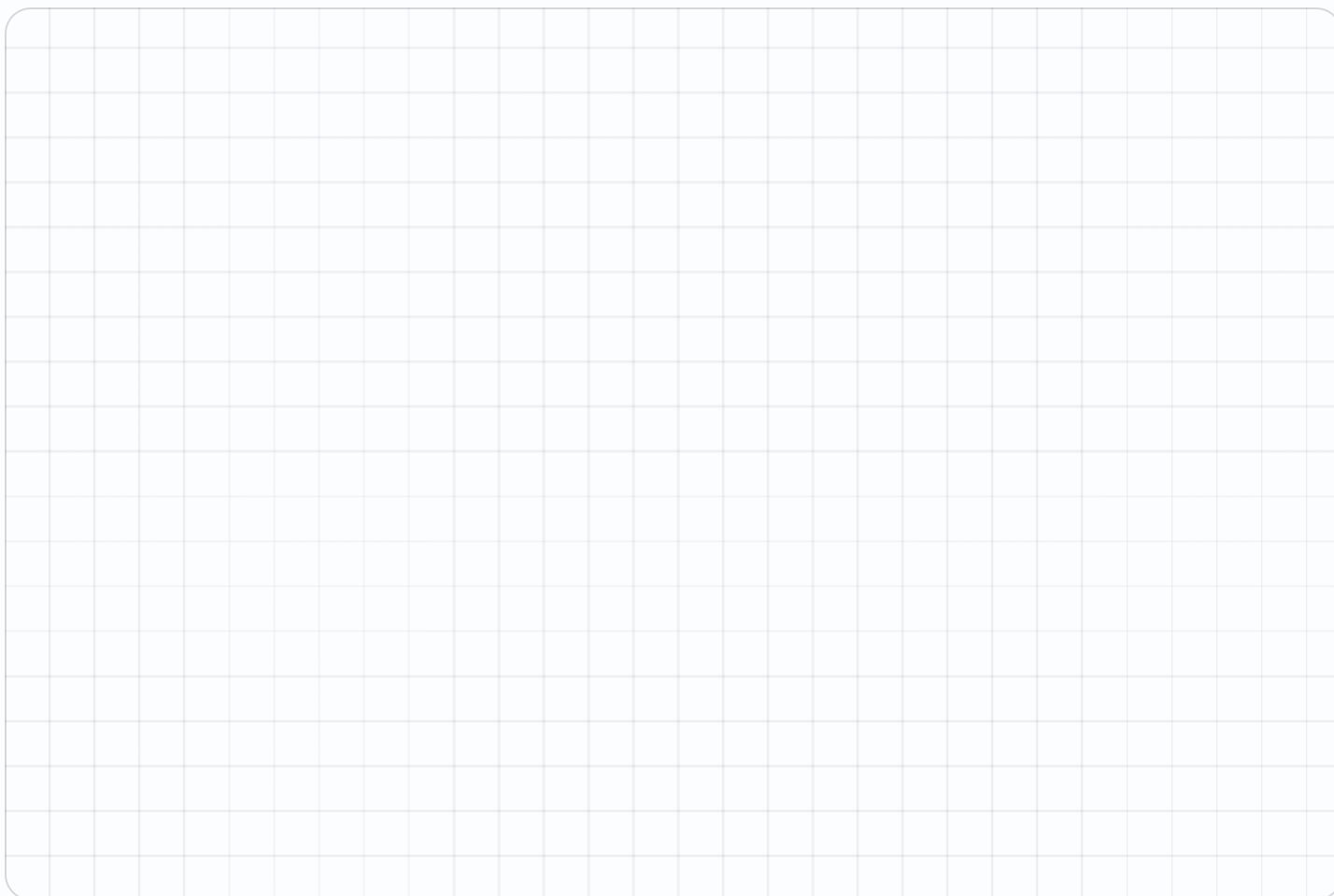
.....

24. A route (2,2) to (9,2) to (9,8) — total blocks

.....

25. The shortest across-and-up route from (1,2) to (10,9) — blocks

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Order Matters

(3,5) and (5,3) are different places. Prove it every time.

Warm-Up 12 ITEMS • RECALL

1 In (3,5), the across value

2 In (5,3), the across value

3 Are (3,5) and (5,3) the same — yes or no

4 In (0,4), the across value

5 In (4,0), the up value

6 A point on the bottom line has which value 0 — type across or up

7 In (4,7), the across value

8 In (7,4), the across value

9 Are (4,7) and (7,4) the same — yes or no

10 In (0,5), the across value

11 In (5,0), the up value

12 A point on the left edge has which value 0 — type across or up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (2,6) and (6,2) — how far apart across

.....

14. Those two points — how far apart up

.....

15. A point with the same across and up as (5,5) — type as a,b

.....

16. Points where across equals up, from (0,0) to (5,5) — how many

.....

17. Swap (7,2) — type the new point as a,b

.....

18. (3,8) and (8,3) — how far apart across

.....

19. A point with the same across and up as (7,7) — type as a,b

.....

20. Points where across equals up, from (0,0) to (7,7) — how many

.....

21. Swap (9,4) — type the new point as a,b

.....

★

Challenge

NOT GRADED · REASON IT OUT

22. Points on the line where across equals up, from 0 to 10 — how many

.....

23. (1,4), (4,1), (1,1) — the area of that triangle

.....

24. Points on the line where across equals up, from 0 to 20 — how many

.....

25. (2,5), (5,2), (2,2) — the area of that triangle

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 Perimeter of a 4 by 3 rectangle

2 Area of a 4 by 3 rectangle

3 Perimeter of a 5 by 4 rectangle

4 Area of a 5 by 4 rectangle

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area 48, one side 6 — the other

.....

6. With 20 m of fence, the biggest area

.....

7. An 8 by 6 with a 2 by 3 removed — area

.....

8. Distance from (2,3) to (7,3)

.....

9. (1,1), (1,5), (6,5), (6,1) — the area

.....

10. Area 72, one side 8 — the other

.....

11. With 40 m of fence, the biggest area

.....

12. A 10 by 8 with a 3 by 4 removed — area

.....

13. Distance from (1,5) to (9,5)

.....

14. (1,1), (1,6), (5,6), (5,1) — the area

.....

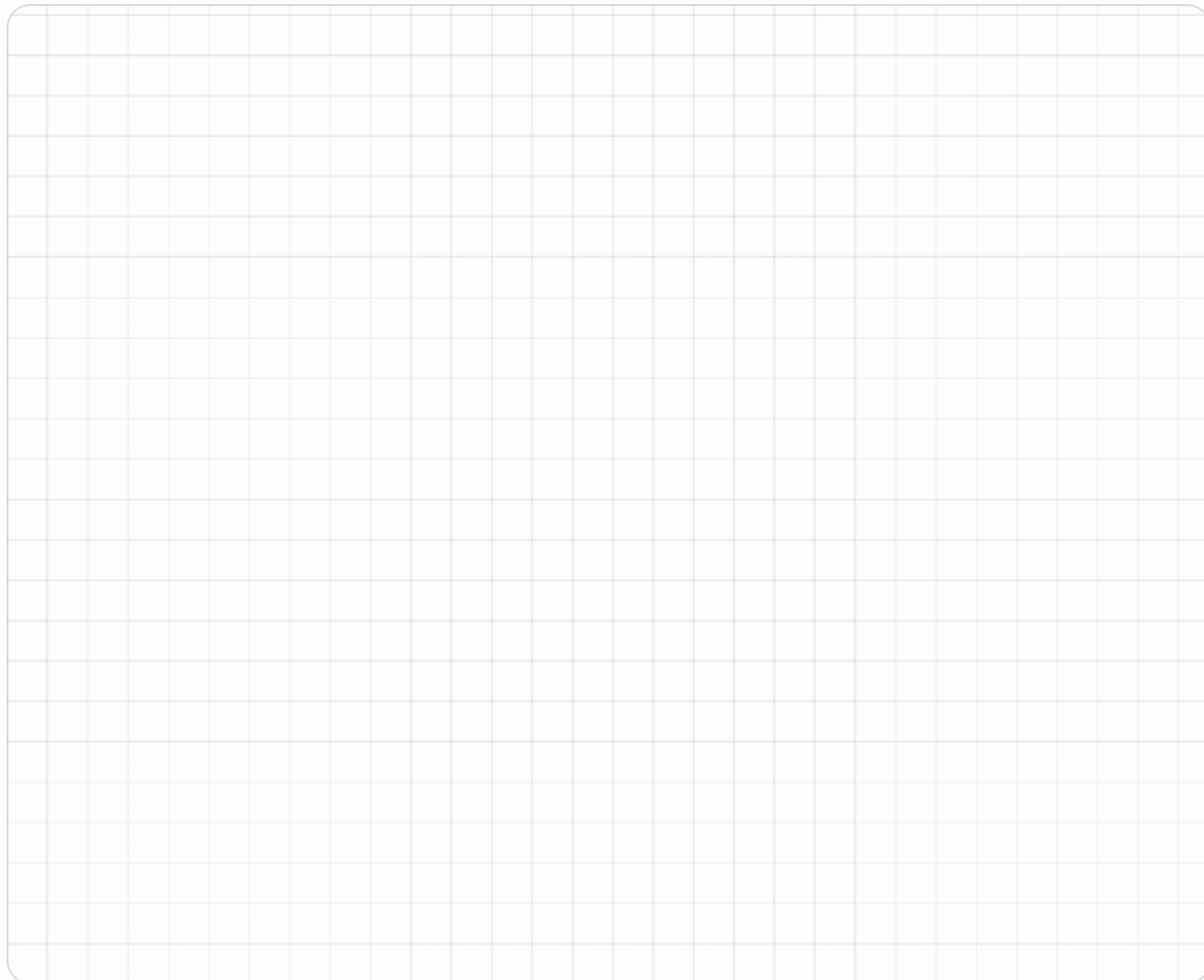
★ Challenge NOT GRADED • REASON IT OUT

15. A rectangle of perimeter 30 with whole sides — how many different ones

.....

16. A rectangle of perimeter 40 with whole sides — how many different ones

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Plotting Points

OUT OF 22

Warm-Up • 1-12 • 3 | 5 | 0,0 | 4,2 | 0,6 | 7 | 5 | 2 | 6,0 | 2,7 | 0,9 | 4

Core • 13-22 • 5,5 | 4,3 | 5 | 7 | 4,0 | 8,4 | 6,5 | 8 | 8 | 6,0

Challenge • 23-26 • 5,4 | 14 | 5,5 | 16

3.2 • Shapes on the Grid

OUT OF 21

Warm-Up • 1-12 • 4 | 3 | 4 | 3 | 4 | 3 | 4 | 5 | 5 | 5 | 4 | 5

Core • 13-21 • 20 | 18 | 15 | square | 6,1 | 20 | 24 | square | 8,2

Challenge • 22-25 • 40 | 25 | 60 | 49

3.3 • Reading a Map

OUT OF 21

Warm-Up • 1-12 • 3,0 | 3,2 | 6,5 | 2,4 | 4 | 7 | 4,0 | 4,3 | 8,6 | 3,7 | 6 | 9

Core • 13-21 • 6,5 | 8 | 6 | 8 | 10 | 8,7 | 9 | 9 | 12

Challenge • 22-25 • 11 | 12 | 13 | 16

3.4 • Order Matters

OUT OF 21

Warm-Up • 1-12 • 3 | 5 | no | 0 | 0 | up | 4 | 7 | no | 0 | 0 | across

Core • 13-21 • 4 | 4 | 5,5 | 6 | 2,7 | 5 | 7,7 | 8 | 4,9

Challenge • 22-25 • 11 | 4.5 | 21 | 4.5

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 14 | 12 | 18 | 20

Core • 5-14 • 8 | 25 | 42 | 5 | 20 | 9 | 100 | 68 | 8 | 20

Challenge • 15-16 • 7 | 10

Lines of Symmetry

Fold it. If both halves land exactly, that fold is a line of symmetry.

Warm-Up 11 ITEMS • RECALL

1 Lines of symmetry in a square

2 In a rectangle that is not a square

3 In a circle — type the word

4 In an equilateral triangle

5 In the letter A

6 In the letter H

7 Lines of symmetry in a rectangle

8 In the letter O — type the word

9 In the letter T

10 In the letter X

11 In a square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. In a regular pentagon
.....

13. In a regular hexagon
.....

14. In an isosceles triangle
.....

15. In a scalene triangle
.....

16. In a rhombus that is not a square
.....

17. In a regular heptagon
.....

18. In a regular nonagon
.....

19. In a kite
.....

20. In a parallelogram that is not a rhombus
.....

21. In a regular decagon
.....

★ Challenge NOT GRADED • REASON IT OUT

22. In a regular octagon

.....

23. In a regular polygon with 12 sides

.....

24. In a regular polygon with 20 sides

.....

25. In a semicircle

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Flips and Turns

Slide, flip, turn. The shape keeps its size and its angles.

✦ Warm-Up 4 ITEMS • RECALL

1 Quarter turns in a full turn

2 Half turns in a full turn

3 Quarter turns in a half turn

4 Half turns in two full turns

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

5. (2,3) flipped over the vertical line through 0 — the across value becomes

.....

6. Slide (2,3) four right — type as a,b

.....

7. (3,4) flipped over the vertical line through 0 — the across value becomes

.....

8. Slide (3,4) five up — type as a,b

.....

 Challenge NOT GRADED · REASON IT OUT

- ### 9. Degrees in a quarter turn

10. Degrees in a half turn

11. Degrees in a full turn

12. Degrees in three quarter turns

13. A square turned 90° looks the same — yes or no

14. Turns of 90° before a square returns to start

15. A rectangle looks the same after how many degrees

16. An equilateral triangle looks the same after how many degrees

17. A regular hexagon — the smallest turn that leaves it unchanged, in degrees

18. Degrees in two right angles

19. Degrees in a three-quarter turn

20. An equilateral triangle turned 120° looks the same — yes or no

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Angles

Right, acute, obtuse — and what they add to.

Warm-Up 12 ITEMS • RECALL

1 Angles in a triangle add to _____

2 A 4 by 3 rectangle joined to a 4 by 2 — total squares _____

3 A 5 by 2 rectangle joined to a 5 by 4 — total squares _____

4 A 6 by 3 rectangle joined to a 6 by 3 — total squares _____

5 A 3 by 5 rectangle joined to a 3 by 5 — total squares _____

6 A 7 by 2 rectangle joined to a 7 by 3 — total squares _____

7 A rectangle 3 squares across and 4 down — squares altogether _____

8 A rectangle 5 squares across and 2 down — squares altogether _____

9 A rectangle 6 squares across and 3 down — squares altogether _____

10 A rectangle 4 squares across and 4 down — squares altogether _____

11 A rectangle 7 squares across and 2 down — squares altogether _____

12 A rectangle 5 squares across and 5 down — squares altogether _____

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Angles in a quadrilateral add to _____

14. $4 \times (3 + 2)$ — work out the bracket first _____

15. 4×3 plus 4×2 _____

16. $5 \times (2 + 4)$ — work out the bracket first _____

17. 5×2 plus 5×4 _____

18. $6 \times (3 + 3)$ — work out the bracket first _____

19. 6×3 plus 6×3 _____

20. A 8 by 3 rectangle and a 8 by 5 joined — total area _____

21. Split a 7 by 6 rectangle into 7 by 5 and 7 by 1 — total area _____

22. A 9 by 3 rectangle and a 9 by 4 joined — total area _____

23. An L shape: a 4 by 3 rectangle and a 2 by 3 rectangle. Total area _____

★

Challenge

NOT GRADED · REASON IT OUT

43. Degrees in a right angle
-
44. Degrees on a straight line
-
45. Degrees round a point
-
46. An angle of 45° — type acute or obtuse
-
47. An angle of 120° — acute or obtuse
-
48. Two angles on a line, one is 60° — the other
-
49. A triangle with 90° and 30° — the third angle
-
50. A quadrilateral with three 90° angles — the fourth
-
51. Three angles round a point, two are 100° — the third
-
52. Each angle of an equilateral triangle
-
53. Each angle of a regular hexagon
-
54. A 10 by 7 rectangle with a 3 by 2 corner removed — area left
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Classifying Shapes

Sides, angles, parallels. Name it from its properties.

Warm-Up 12 ITEMS • RECALL

1 Sides on a pentagon

2 Sides on a hexagon

3 Sides on an octagon

4 Sides on a quadrilateral

5 Equal sides on an equilateral triangle

6 Sides on a triangle

7 Sides on a heptagon

8 Sides on a nonagon

9 Sides on a decagon

10 Sides on a rhombus

11 Equal sides on an isosceles triangle

12 Sides on a square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is every square a rectangle — yes or no

.....

14. Is every rectangle a square

.....

15. Pairs of parallel sides in a parallelogram

.....

16. Pairs of parallel sides in a trapezoid

.....

17. Is every rhombus a parallelogram — yes or no

.....

18. Is every parallelogram a rhombus

.....

19. Pairs of parallel sides in a rectangle

.....

20. Right angles in a rectangle

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Tessellation

Which shapes tile a floor with no gaps? Test three and explain.

✦ Warm-Up 4 ITEMS • RECALL

1 Do squares tile with no gaps — yes or no

2 Do circles tile with no gaps

3 Do rectangles tile with no gaps — yes or no

4 Do regular octagons tile alone

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Do equilateral triangles tile — yes or no

.....

6. Do regular hexagons tile

.....

7. Do parallelograms tile — yes or no

.....

8. Do regular heptagons tile

.....

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Lines of Symmetry

OUT OF 21

Warm-Up • 1-11 • 4 | 2 | infinite | 3 | 1 | 2 | 2 | infinite | 1 | 2 | 4

Core • 12-21 • 5 | 6 | 1 | 0 | 2 | 7 | 9 | 1 | 0 | 10

Challenge • 22-25 • 8 | 12 | 20 | 1

4.2 • Flips and Turns

OUT OF 8

Warm-Up • 1-4 • 4 | 2 | 2 | 4

Core • 5-8 • -2 | 6,3 | -3 | 3,9

Challenge • 9-20 • 90 | 180 | 360 | 270 | yes | 4 | 180 | 120 | 60 | 180 | 270 |
yes

4.3 • Angles

OUT OF 23

Warm-Up • 1-12 • 180 | 20 | 30 | 36 | 30 | 35 | 12 | 10 | 18 | 16 | 14 | 25

Core • 13-23 • 360 | 20 | 20 | 30 | 30 | 36 | 36 | 64 | 42 | 63 | 18

Challenge • 43-54 • 90 | 180 | 360 | acute | obtuse | 120 | 60 | 90 | 160 | 60 |
120 | 64

4.4 • Classifying Shapes

OUT OF 20

Warm-Up • 1-12 • 5 | 6 | 8 | 4 | 3 | 3 | 7 | 9 | 10 | 4 | 2 | 4

Core • 13-20 • yes | no | 2 | 1 | yes | no | 2 | 4

Challenge • 21-26 • right | square | scalene | acute | rhombus | isosceles

Fri • Tessellation

OUT OF 8

Warm-Up • 1-4 • yes | no | yes | no

Core • 5-8 • yes | yes | yes | no

Challenge • 9-14 • no | 360 | 3 | 4 | 6 | 36

Map a Planet

An invented world on the coordinate plane, with real distances.

Warm-Up 11 ITEMS • RECALL

1 A harbour at (2,3) — the across value

2 A peak at (7,7) — the up value

3 Distance from (2,3) to (6,3)

4 Distance from (5,1) to (5,6)

5 The origin — type as a,b

6 3 across, 8 up — type as a,b

7 A dock at (3,4) — the across value

8 A peak at (8,9) — the up value

9 Distance from (3,4) to (8,4)

10 Distance from (6,2) to (6,9)

11 5 across, 5 up — type as a,b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Sailing (1,1) to (9,1) then to (9,5) — total distance

.....

13. A square island (2,2) to (8,8) — its area

.....

14. That island's perimeter

.....

15. Halfway between (0,0) and (10,4) — type as a,b

.....

16. A route (0,0) to (4,0) to (4,7) — distance

.....

17. Sailing (2,2) to (10,2) then to (10,7) — total distance

.....

18. A square island (1,1) to (9,9) — its area

.....

19. Halfway between (0,0) and (12,8) — type as a,b

.....

20. A route (1,1) to (6,1) to (6,9) — distance

.....

★ Challenge NOT GRADED • REASON IT OUT

21. A rectangular sea from (1,1) to (11,6) — its area

.....

22. Ten landmarks each 3 units apart in a line — total length

.....

23. A rectangular sea from (2,2) to (14,8) — its area

.....

24. Twelve landmarks each 4 units apart in a line — total length

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Sailing Directions

Somebody else has to follow them. Ambiguity is the enemy.

Warm-Up 11 ITEMS • RECALL

1 From (0,0) move 5 across — type as a,b

2 Then 3 up — type as a,b

3 Total distance travelled

4 From (4,4) move 2 up — type as a,b

5 From (4,4) move 2 across — type as a,b

6 Blocks from (0,0) to (3,4) across-and-up

7 From (0,0) move 7 across — type as a,b

8 Then 2 up — type as a,b

9 From (5,5) move 3 up — type as a,b

10 From (5,5) move 3 across — type as a,b

11 Blocks from (0,0) to (5,6) across-and-up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. (1,1) to (1,7) to (5,7) — total distance

13. (2,2) to (8,2) to (8,9) — total distance

14. A round trip (0,0) to (6,0) and back — distance

15. (3,3) to (3,10) — distance

16. A square patrol of side 5 — total distance

17. (2,1) to (2,9) to (7,9) — total distance

18. (3,3) to (11,3) to (11,10) — total distance

19. A round trip (0,0) to (8,0) and back — distance

20. (4,4) to (4,12) — distance

21. A square patrol of side 7 — total distance

★ Challenge NOT GRADED • REASON IT OUT

22. A patrol (1,1),(7,1),(7,5),(1,5) and back to start — total distance

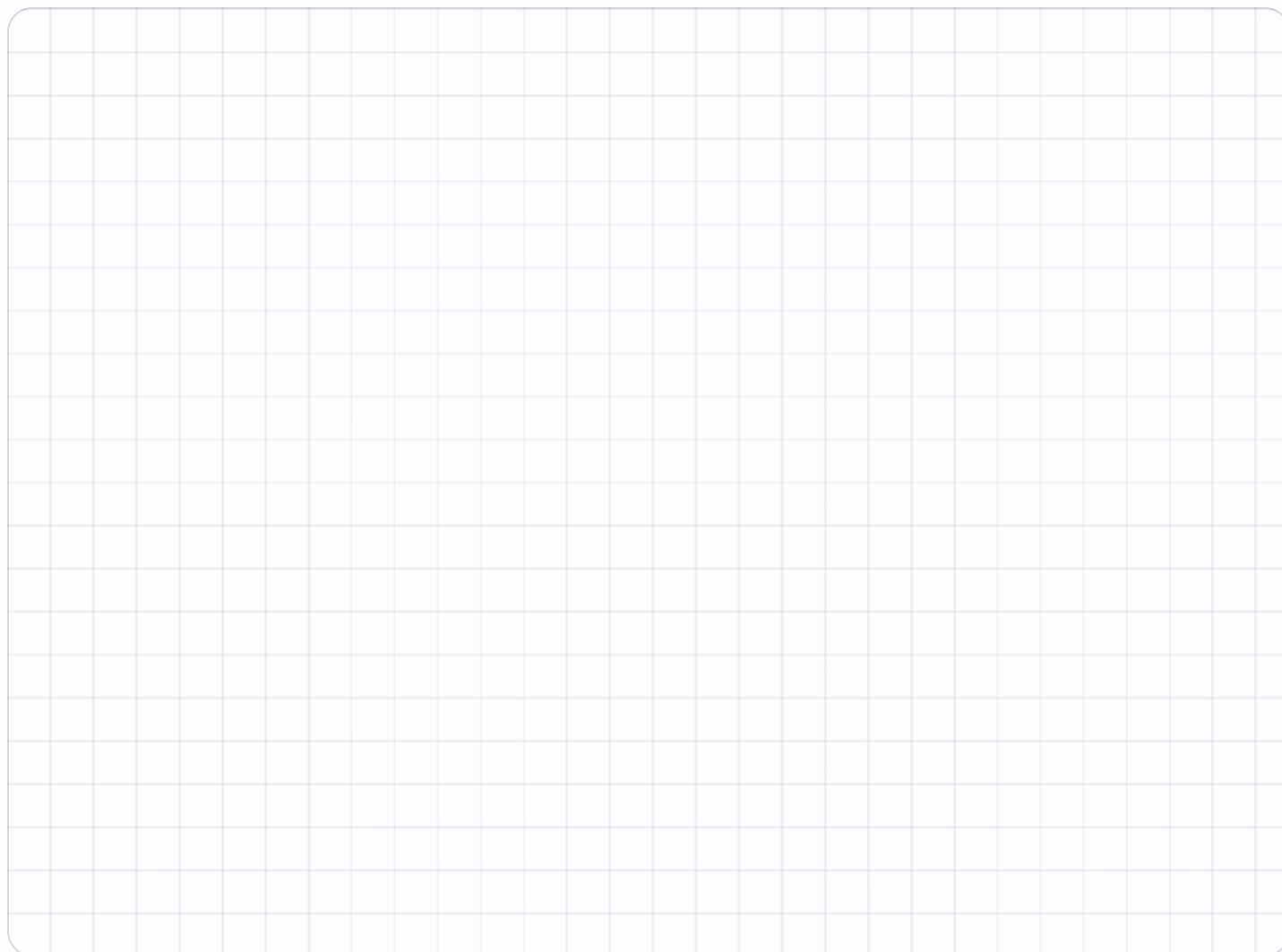
.....

23. The area enclosed by that patrol

.....

24. A patrol (2,2),(9,2),(9,7),(2,7) and back — total distance

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mission Review

Everything from five weeks, mixed together.

Warm-Up 10 ITEMS • RECALL

1 Perimeter of a 5 by 5 square

2 Area of a 5 by 5 square

3 Lines of symmetry in a square

4 In (3,5), the up value

5 Angles in a triangle add to

6 Perimeter of a 6 by 6 square

7 Area of a 6 by 6 square

8 Lines of symmetry in an equilateral triangle

9 In (6,2), the up value

10 Angles in a quadrilateral add to

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Area 96 with one side 12 — the other
.....

12. With 24 m of fence, the biggest area
.....

13. (2,2), (2,7), (5,7), (5,2) — the area
.....

14. Lines of symmetry in a regular hexagon
.....

15. Area 144 with one side 16 — the other
.....

16. With 32 m of fence, the biggest area
.....

17. (3,2), (3,8), (7,8), (7,2) — the area
.....

18. Lines of symmetry in a regular octagon
.....

 Challenge NOT GRADED • REASON IT OUT

19. Degrees in a right angle
20. A triangle with 90° and 30° — the third angle
21. A 10 by 10 with a 4 by 4 removed — area
22. Each angle of a regular hexagon
23. Degrees on a straight line
24. A triangle with 50° and 60° — the third angle
25. A 14 by 14 with a 6 by 6 removed — area
26. Each angle of a regular pentagon

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 10 ITEMS • RECALL

1 Perimeter of a 6 by 6 square

2 Area of a 8 by 3 rectangle

3 Sides on a hexagon

4 Distance from (2,3) to (7,3)

5 Lines of symmetry in a rectangle

6 Perimeter of a 8 by 8 square

7 Area of a 9 by 4 rectangle

8 Sides on a decagon

9 Distance from (1,5) to (9,5)

10 Lines of symmetry in a square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. An 8 by 6 with a 2 by 3 removed — area

.....

12. A square of perimeter 36 — one side

.....

13. (1,1), (1,5), (6,5), (6,1) — the perimeter

.....

14. Is every square a rectangle — yes or no

.....

15. A 10 by 8 with a 3 by 4 removed — area

.....

16. A square of perimeter 48 — one side

.....

17. (1,1), (1,6), (5,6), (5,1) — the perimeter

.....

18. Is every rhombus a parallelogram — yes or no

.....

Mission 06 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 Perimeter of a 10 by 2 rectangle

2 Area of a 10 by 2 rectangle

3 Perimeter of a 12 by 1 rectangle

4 Area of a 12 by 1 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area of a 12 by 8 rectangle

.....

6. A rectangle of perimeter 20, one side 6 — the other

.....

7. With 20 m of fence, the biggest area

.....

8. A 12 by 5 with a 3 by 5 removed — area

.....

9. Distance from (4,1) to (4,8)

.....

10. (1,1), (1,5), (6,5), (6,1) — the area

.....

11. Lines of symmetry in a regular pentagon

.....

12. Area of a 16 by 9 rectangle

.....

13. A rectangle of perimeter 26, one side 8 — the other

.....

14. With 40 m of fence, the biggest area

.....

15. A 16 by 7 with a 5 by 7 removed — area

.....

16. Distance from (7,2) to (7,10)

.....

17. (3,2), (3,8), (7,8), (7,2) — the area

.....

★ Challenge NOT GRADED • REASON IT OUT

19. A triangle with 90° and 30° — the third angle

.....

20. A 20 by 15 room with a 4 by 5 alcove added — total area

.....

21. A patrol (1,1),(7,1),(7,5),(1,5) — the area enclosed

.....

22. A triangle with 90° and 45° — the third angle

.....

23. A 30 by 20 room with a 6 by 5 alcove added — total area

.....

24. A patrol (2,2),(9,2),(9,7),(2,7) — the area enclosed

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Map a Planet

OUT OF 20

Warm-Up • 1-11 • 2 | 7 | 4 | 5 | 0,0 | 3,8 | 3 | 9 | 5 | 7 | 5,5

Core • 12-20 • 12 | 36 | 24 | 5,2 | 11 | 13 | 64 | 6,4 | 13

Challenge • 21-24 • 50 | 27 | 72 | 44

5.2 • Sailing Directions

OUT OF 21

Warm-Up • 1-11 • 5,0 | 5,3 | 8 | 4,6 | 6,4 | 7 | 7,0 | 7,2 | 5,8 | 8,5 | 11

Core • 12-21 • 10 | 13 | 12 | 7 | 20 | 13 | 15 | 16 | 8 | 28

Challenge • 22-24 • 20 | 24 | 24

5.3 • Mission Review

OUT OF 18

Warm-Up • 1-10 • 20 | 25 | 4 | 5 | 180 | 24 | 36 | 3 | 2 | 360

Core • 11-18 • 8 | 36 | 15 | 6 | 9 | 64 | 24 | 8

Challenge • 19-26 • 90 | 60 | 84 | 120 | 180 | 70 | 160 | 108

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 24 | 24 | 360 | 6 | 5 | 2 | 32 | 36 | 360 | 10 | 8 | 4

Core • 13-22 • 42 | 9 | 120 | 18 | yes | 68 | 12 | 145 | 18 | yes

Challenge • 23-26 • 100 | 60 | 225 | 45

Fri • Mission 06 Test

OUT OF 20

Warm-Up • 1-4 • 24 | 20 | 26 | 12

Core • 5-20 • 96 | 4 | 25 | 45 | 7 | 20 | 5 | 60 | 144 | 5 | 100 | 77 | 8 |
24 | 7 | 45

Challenge • 21-24 • 320 | 24 | 630 | 35